

Emergency Panic Tracker – ESP8266 Code

```
#include <Wire.h>
#include <Adafruit_MPU6050.h>
#include <Adafruit_Sensor.h>

#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>

// =====
// WIFI
// =====

const char* WIFI_SSID = "YOUR_WIFI_NAME";
const char* WIFI_PASSWORD = "YOUR_WIFI_PASSWORD";

// =====
// REPLIT SERVER
// =====

// Example:
// https://your-project-name.yourusername.repl.co/api/alert

const char* SERVER_URL = "YOUR_REPLIT_URL";

// =====
// PIN CONNECTIONS
// =====

#define PANIC_BUTTON D5
#define LED_PIN D6
#define BUZZER_PIN D7

// =====
// MPU6050
// =====

Adafruit_MPU6050 mpu;

// =====
// VARIABLES
// =====

int alertCount = 0;

unsigned long lastPress = 0;

const unsigned long debounceTime = 1000;

// =====
// WIFI CONNECTION
// =====

void connectWiFi()
{
  Serial.println();
  Serial.println("Connecting to WiFi...");

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);

  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);

    Serial.print(".");
  }

  Serial.println();
  Serial.println("WiFi connected!");

  Serial.print("ESP8266 IP address: ");

  Serial.println(WiFi.localIP());
}

// =====
// SEND DATA TO REPLIT
// =====

void sendEmergencyToReplit(
  float X,
  float Y,
  float Z,
  float temperature
)
{
  if (WiFi.status() != WL_CONNECTED)
  {
    Serial.println("WiFi disconnected!");
    connectWiFi();
  }
}
```

```

WiFiClient client;

HTTPClient http;

Serial.println();
Serial.println("Sending emergency data to Replit...");

http.begin(client, SERVER_URL);

http.addHeader(
  "Content-Type",
  "application/json"
);

// Create JSON data

String jsonData = "{";

jsonData += "\"alert\":true,";

jsonData += "\"alertCount\":";
jsonData += String(alertCount);
jsonData += ",";

jsonData += "\"accelerationX\":";
jsonData += String(X, 2);
jsonData += ",";

jsonData += "\"accelerationY\":";
jsonData += String(Y, 2);
jsonData += ",";

jsonData += "\"accelerationZ\":";
jsonData += String(Z, 2);
jsonData += ",";

jsonData += "\"temperature\":";
jsonData += String(temperature, 2);

jsonData += "}";

Serial.println("Data:");

Serial.println(jsonData);

// Send POST request

int httpResponseCode =
  http.POST(jsonData);

Serial.print("Server response: ");

Serial.println(httpResponseCode);

if (httpResponseCode > 0)
{
  String response =
    http.getString();

  Serial.println("Replit response:");

  Serial.println(response);
}
else
{
  Serial.println(
    "ERROR: Could not contact Replit!"
  );
}

http.end();
}

// =====
// SETUP
// =====

void setup()
{
  Serial.begin(115200);

  // -----
  // SET PINS
  // -----

  pinMode(
    PANIC_BUTTON,
    INPUT_PULLUP
  );

  pinMode(
    LED_PIN,
    OUTPUT
  );

```

```

);

pinMode(
  BUZZER_PIN,
  OUTPUT
);

digitalWrite(
  LED_PIN,
  LOW
);

digitalWrite(
  BUZZER_PIN,
  LOW
);

// -----
// START I2C
// -----

Wire.begin(D2, D1);

Serial.println();

Serial.println(
  "=====
");

Serial.println(
  " EMERGENCY PANIC TRACKER"
);

Serial.println(
  "=====
");

// -----
// START MPU6050
// -----

if (!mpu.begin())
{
  Serial.println(
    "ERROR: MPU6050 NOT FOUND!"
  );

  while (true)
  {
    digitalWrite(
      LED_PIN,
      HIGH
    );

    delay(300);

    digitalWrite(
      LED_PIN,
      LOW
    );

    delay(300);
  }
}

Serial.println(
  "MPU6050 connected!"
);

// -----
// CONFIGURE MPU6050
// -----

mpu.setAccelerometerRange(
  MPU6050_RANGE_8_G
);

mpu.setGyroRange(
  MPU6050_RANGE_500_DEG
);

mpu.setFilterBandwidth(
  MPU6050_BAND_21_HZ
);

// -----
// CONNECT WIFI
// -----

connectWiFi();

// -----
// SYSTEM READY

```

```

// -----
Serial.println();

Serial.println(
    "=====
");

Serial.println(
    " SYSTEM READY"
);

Serial.println(
    " Press the panic button..."
);

Serial.println(
    "=====
");
}

// =====
// PANIC ALERT
// =====

void panicAlert()
{
    alertCount++;

    Serial.println();

    Serial.println(
        "=====
    );

    Serial.println(
        "      EMERGENCY ALERT"
    );

    Serial.println(
        "=====
    );

    // -----
    // TURN ON ALARM
    // -----

    digitalWrite(
        LED_PIN,
        HIGH
    );

    digitalWrite(
        BUZZER_PIN,
        HIGH
    );

    // -----
    // READ MPU6050
    // -----

    sensors_event_t acceleration;
    sensors_event_t gyro;
    sensors_event_t temperature;

    mpu.getEvent(
        &acceleration,
        &gyro,
        &temperature
    );

    // -----
    // STORE VALUES
    // -----

    float X =
        acceleration.acceleration.x;

    float Y =
        acceleration.acceleration.y;

    float Z =
        acceleration.acceleration.z;

    float temp =
        temperature.temperature;

    // -----
    // DISPLAY VALUES
    // -----
    Serial.print(

```

```

    "Acceleration X: "
);

Serial.print(X);

Serial.println(
    " m/s^2"
);

Serial.print(
    "Acceleration Y: "
);

Serial.print(Y);

Serial.println(
    " m/s^2"
);

Serial.print(
    "Acceleration Z: "
);

Serial.print(Z);

Serial.println(
    " m/s^2"
);

Serial.print(
    "Temperature: "
);

Serial.print(temp);

Serial.println(
    " C"
);

Serial.print(
    "Alert Count: "
);

Serial.println(
    alertCount
);

// -----
// SEND TO REPLIT
// -----

sendEmergencyToReplit(
    X,
    Y,
    Z,
    temp
);

Serial.println();

Serial.println(
    "EMERGENCY SIGNAL SENT!"
);

// -----
// ALARM FOR 3 SECONDS
// -----

unsigned long startTime =
    millis();

while (
    millis() - startTime < 3000
)
{
    delay(10);
}

// -----
// TURN OFF ALARM
// -----

digitalWrite(
    BUZZER_PIN,
    LOW
);

digitalWrite(
    LED_PIN,
    LOW
);

Serial.println(

```

```

    "Alarm stopped."
);

Serial.println();
}

// =====
// MAIN LOOP
// =====

void loop()
{
    // -----
    // CHECK WIFI
    // -----

    if (WiFi.status() != WL_CONNECTED)
    {
        connectWiFi();
    }

    // -----
    // CHECK PANIC BUTTON
    // -----

    if (
        digitalRead(PANIC_BUTTON)
        == LOW
    )
    {
        unsigned long currentTime =
            millis();

        if (
            currentTime - lastPress
            > debounceTime
        )
        {
            lastPress =
                currentTime;

            panicAlert();
        }
    }

    delay(50);
}

```